

Transmission and Substation Project Cost Estimation Guide For MTEP 2018

Disclaimer

This document is prepared for informational purposes only to support MISO planning staff in developing cost estimates and deriving benefit-to-cost ratios for solutions proposed for inclusion in the MISO Transmission Expansion Plan (MTEP). MISO's cost estimation process is based on staff experience, vendor consultation, industry practice, and stakeholder feedback. MISO makes every effort to develop its cost estimates from the most accurate and appropriate assumptions and information available at that time. However, MISO cannot and does not guarantee the accuracy of information, assumptions, judgments, or opinions contained herein or derived therefrom. MISO may revise or terminate this document at any time at its discretion without notice.

1. Table of Contents

1.	TABLE OF CONTENTS.....	3
2.	MISO'S ROLE PROVIDING COST ESTIMATES	4
3.	MISO'S COST ESTIMATION APPROACH	5
3.1	TRANSMISSION LINES	6
3.1.1	<i>Land and Right-of-Way.....</i>	7
3.1.2	<i>Structures.....</i>	7
3.1.3	<i>Conductor, Shieldwire, and Accessories.....</i>	8
3.1.4	<i>Overhead and professional services</i>	9
3.2	SUBSTATIONS	9
3.2.1	<i>Land and site work.....</i>	10
3.2.2	<i>Equipment.....</i>	11
3.2.3	<i>Protection and Control.....</i>	12
3.2.4	<i>Overhead and professional services</i>	13
3.3	INDUSTRY PRACTICES FOR PROJECT COST ESTIMATION	14
4.	MISO'S COST ESTIMATION DATA	15
4.1	TRANSMISSION LINE COST DATA	15
4.1.1	<i>Land and Right-of-Way.....</i>	15
4.1.2	<i>Structures.....</i>	18
4.1.3	<i>Conductor, Shieldwire, and Accessories.....</i>	33
4.1.4	<i>Overhead and professional services</i>	37
4.2	SUBSTATION COST DATA.....	38
4.2.1	<i>Site Work</i>	38
4.2.2	<i>Equipment.....</i>	41
4.2.3	<i>Protection and Control.....</i>	48
4.2.4	<i>Overhead and professional services</i>	51
5.	PROJECT IMPLEMENTATION COST	52
6.	EXPLORATORY COST ESTIMATES.....	53
6.1	TRANSMISSION LINE EXPLORATORY COSTS	53
6.2	SUBSTATION EXPLORATORY COSTS	55
6.3	ADD ONE POWER TRANSFORMER TO AN EXISTING SUBSTATION SITE	56

2. MISO's role providing cost estimates

The MISO transmission planning process focuses on maximizing value to consumers while minimizing the total energy, capacity and transmission costs. As a part of this process, MISO identifies essential transmission projects that will improve the reliability and efficiency of energy delivery in the region. Those projects are included in the MISO Transmission Expansion Plan (MTEP), an annual publication that is collaboration between MISO planning staff and stakeholders.

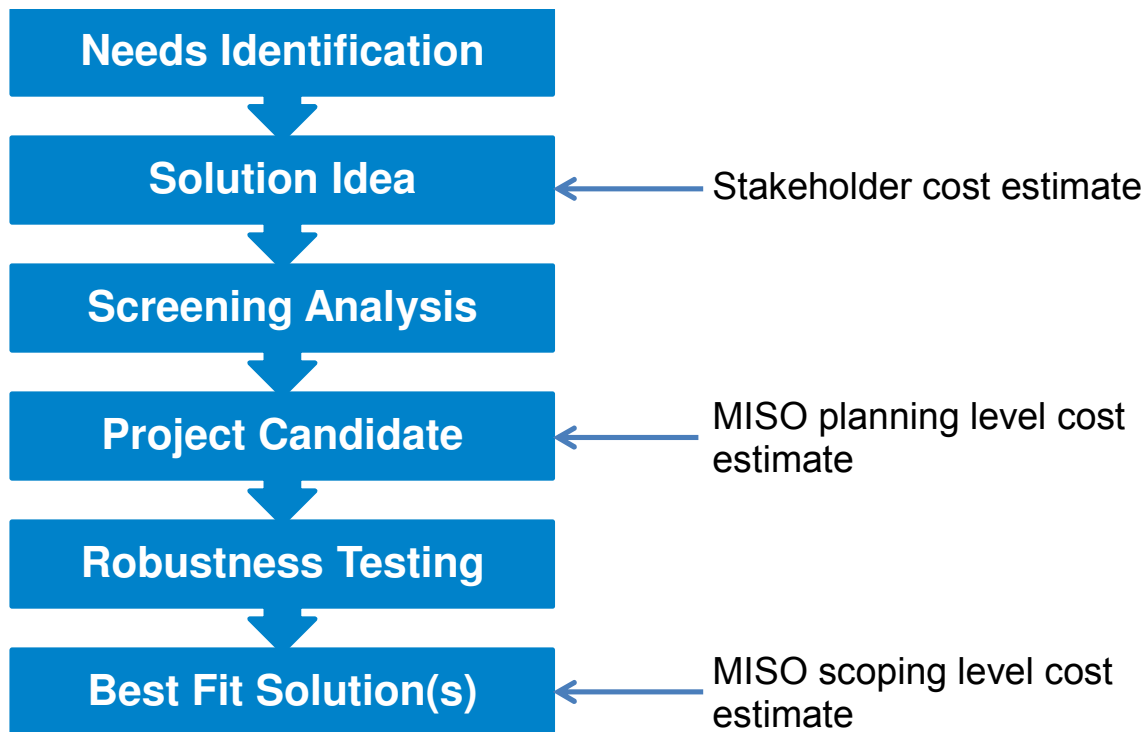
Historically, Transmission Owners provided cost estimates for projects that fell within their service territory during development of the MISO's planning process (MTEP). The introduction of FERC Order 1000, and the elimination of incumbent Transmission Owner's federal rights of first refusal (ROFR) created competition for projects that meet certain requirements. MISO facilitates this competition and utilizes a Competitive Transmission Process (CTP) to select qualified entities to construct, own, operate, maintain, repair, and restore eligible new transmission and substation facilities located within all or parts of the 15 U.S. states in the MISO footprint, as allowed by applicable state law. Project cost is one factor on which Transmission Owners and Developers compete in MISO's Competitive Transmission Process. The Planning Advisory Committee (PAC) sought MISO's involvement in creating cost estimates for potential projects eligible for competitive solicitation.

MISO's transmission and substation project cost estimation guide for MTEP 2018 describes the approach and cost data that MISO uses in developing cost estimates. This document's assumptions and cost data are reviewed yearly with stakeholders.

All cost estimate data in this document are in 2018 Dollars.

3. MISO's Cost Estimation Approach

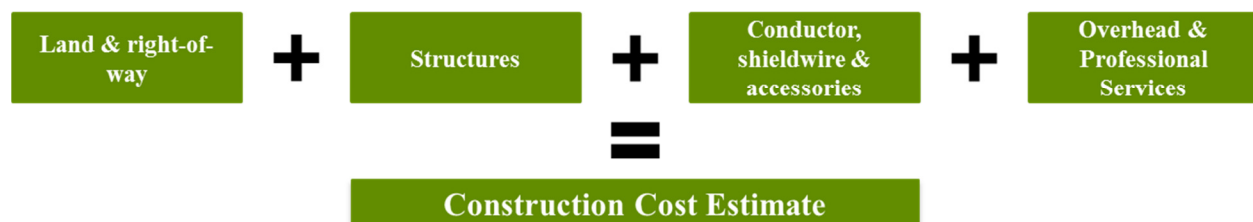
In the yearly MTEP process, MISO performs Market Congestion Planning Studies (MCPS) ([https://www.misoenergy.org/planning/transmission-studies-and-reports/#/report-study-analysistype|Economic Studies](https://www.misoenergy.org/planning/transmission-studies-and-reports/#/report-study-analysistype|Economic%20Studies)). The goal of the MCPS is to develop transmission plans that offer MISO customers better access to the lowest electric energy costs through the markets. As an integral part of MISO's value-based planning, the MCPS looks to develop the most robust transmission upgrades that offer the highest future value under a variety of both current and projected system scenarios. MISO develops project cost estimates specifically for facilities in the MCPS. MISO's Tariff requires a benefit-to-cost ratio (financial benefits/project estimated costs) equal to or greater than 1.0 for Multi-Value Projects and 1.25 for Market Efficiency Projects. The flow chart below gives of an overview of the MCPS planning process at a high level:



MISO uses two levels of cost estimation – planning level cost estimates, and scoping level cost estimates. Planning level cost estimates are project cost estimates based upon general assumptions for the project area. Planning level cost estimates are used as part of the screening process for MISO's MCPS to identify potential projects for further study. Scoping level cost estimates are a more detailed estimate based on the specific site. Scoping level cost estimates are used to justify the business case as to whether a potential project shall be recommended to the MISO Board of Directors for their approval. When MISO creates its scoping level cost estimate for a potential project, MISO may adjust any of its cost estimate assumptions and/or any of its unit costs as necessary for that specific potential project. MISO's cost estimation assumptions are not an indication or a direction for how any particular project should be designed or built.

3.1 Transmission Lines

Transmission line cost estimates are sub-divided into four categories: land and right-of-way; structures; conductor, shieldwire and accessories; overhead and professional services. MISO's quantity assumptions are described in section 3 of this guide, and MISO's cost data for those items are described in section 4 of this guide. The diagram below shows the four categories of a transmission line cost estimate:



The line length for a transmission line is a consideration for determining its cost estimate for a potential project. For planning level cost estimates, the line length is determined by the straight line distance between the two substations plus a 30% line length adder. This 30% line length adder is intended to account for routing constraints that will be determined upon further development of the potential transmission line project. For scoping level cost estimates, the line length is determined by a MISO-created proxy route based upon a desktop study. For new potential projects, MISO considers new right-of-way, and for retrofit/re-conductor projects, MISO considers that existing right-of-way is adequate. MISO does not share its assumed proxy route information with stakeholders, as the route could be perceived as a MISO endorsed/preferred route. MISO's proxy route is merely an instrument to support the MISO's transmission line cost estimate. MISO utilizes Google Earth to determine route length, and terrain types encountered.

3.1.1 Land and Right-of-Way

Land and right-of-way costs are all the costs required to acquire and prepare the land area for new potential transmission line projects. Land costs are based upon the acreage of land that the new transmission line would traverse. The total land affected is the length of the transmission line multiplied by the right-of-way width of the line. MISO's assumptions for right-of-way width are below. The right-of-way widths that MISO considers are indicative of right-of-way widths for transmission lines in each voltage class. Different project conditions in different locations may have a wider or narrower right-of-way width than the indicative value MISO assumes. MISO selects a right-of-way width for the purposes of creating its cost estimate.

Single circuit and double circuit							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Right-of-way width (feet)	80	90	95	100	125	175	200

Right-of-way acquisition costs, terrain costs, and grading costs are based on the length of the new transmission line, and the topography encountered along the potential transmission line route. MISO will assume certain project specific mitigation costs when necessary. Mobilization/Demobilization unit costs are based on the voltage class of the transmission line.

3.1.2 Structures

Structure costs are all the costs required to procure and install structures for new potential transmission line projects. The structure types and quantities per mile for new potential transmission line projects are determined by its voltage level. For planning level project cost estimates, MISO uses the structure counts per mile as shown below. For scoping level cost estimates, MISO uses structure counts as dictated by its proxy route and average span length from tables below.

Structures per miles - single circuit							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Tangent structures per mile	9.0	8.5	8.0	7.0	5.0	4.5	3.0
Small angle structures per mile	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Large angle structures per mile	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Strain structures per mile	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Deadend structures per mile	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Total Structures per miles	10.5	10.0	9.5	8.5	6.5	6.0	4.5

Structures per miles - double circuit							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Tangent structures per mile	9.5	9.0	8.5	7.5	7.0	6.0	5.0
Small angle structures per mile	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Large angle structures per mile	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Strain structures per mile	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Deadend structures per mile	0.25	0.25	0.25	0.25	0.25	0.25	0.25
Total structures per mile	11.0	10.5	10.0	9.0	8.5	7.5	6.5

Structure costs are also based upon the structure type used. For each voltage class, MISO has a primary structure type used, and an alternative structure type used as shown in the table below:

Single circuit and double circuit							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Primary structure type	Steel pole	Steel pole	Steel pole	Steel pole	Steel pole	Steel pole	Steel lattice tower
Alternative structure type (For information only)	Steel lattice tower	Steel lattice tower	Steel lattice tower	Steel lattice tower	Steel lattice tower	Steel lattice tower	Steel pole

3.1.3 Conductor, Shieldwire, and Accessories

Conductor, shieldwire, and accessory costs are all the costs required to procure and install the conductor, shieldwire and accessories required for new potential transmission line projects. Conductor costs are based upon the conductor selected and the length of the transmission line. Conductor type and size are based on the information contained in solution idea submission form and are based on economic planning model considerations for the required ampacity. MISO uses its Business Practice Manual 029 to assign appropriate conductor type and size for a project in lieu of necessary conductor information in the solution idea form. The table below shows MISO's conductor selection:

Conductor selection per circuit							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Conductor size and type	1-477kcmil ACSS	1-795kcmil ACSS	1-795kcmil ACSS	1-795kcmil ACSS	1-795kcmil ACSS	2-795kcmil ACSS	3-954kcmil ACSR

Some solution ideas involve re-conductoring or upgrading existing conductor size to allow more power transfer by increasing ampacity of the existing circuit. In providing cost estimates for re-conductoring project scope, MISO assumes that the existing structures, insulators and hardware are adequate to support the new conductor size and configuration and discusses this assumption with the Transmission Owner. The costs of new conductor and installation are considered for the estimate of the retrofit projects.

Unless otherwise specified by the solution idea, MISO assumes one OPGW and one steel shieldwire per transmission circuit.

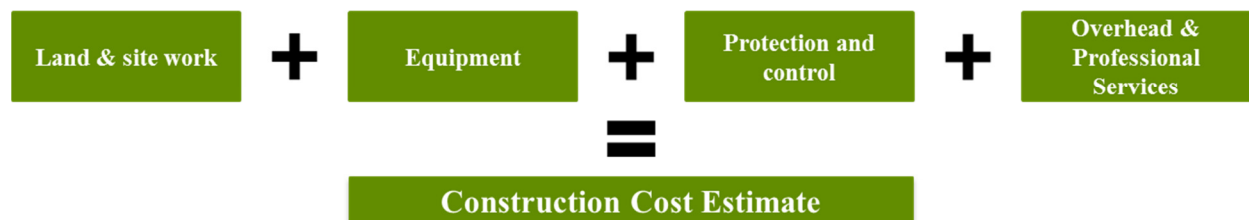
MISO assumes conductor and shieldwire length adder of 4% for sag and wastage per conductor, OPGW, and shieldwire.

3.1.4 Overhead and professional services

Overhead and professional services encompass all the indirect overhead costs (e.g. computers, trucks) and all the professional services (e.g. surveying, drafting, engineering, project management, testing and commissioning) required for potential projects. This cost adder is to account for all the added costs that are spread out over the entire project and are not easily quantifiable by individual items.

3.2 Substations

Substation cost estimates are sub-divided in to four cost categories: land and site work; equipment; protection and control; overhead and professional services. MISO's quantity assumptions are described in section 3 of this guide, and MISO's cost data for those items are described in section 4 of this guide. The diagram below shows the four categories of a substation cost estimate:



MISO provides cost estimates for both substation upgrades and for new substation sites. For planning level cost estimates, MISO assumes size (acreage) requirements and equipment quantities based on general assumptions for the project area. Both the size of the substation facilities and the equipment quantities are dependent upon the voltage class of the facility and the number of new line/transformer positions being considered. For scoping level cost estimates

that are upgrades of existing substations, MISO discusses its scope of work assumptions with the existing substation owner. If the substation is a new facility, MISO follows requirements in its Business Practice Manual 029 (BPM-029).

3.2.1 Land and site work

Land and site work costs are all the costs required to procure and install the substation general facilities. Mobilization/Demobilization unit costs are based on whether a substation site is new, or existing. Land, clearing and grubbing, grading, and site/fence/grounding unit costs for the substation are based on the acreage required to complete the substation work. For a planning level estimate, MISO assumes the land required as listed in the tables below. For a scoping level cost estimate, MISO discusses the land assumptions with the substation owner where the new potential project is a substation upgrade.

Substation upgrade - land required per line position (acre)							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Land required (acre)	0.38	0.44	0.50	0.63	1.00	2.00	5.00

Where the substation site is new, the land required shown below is where the new site has four new line positions, and a breaker-and-a-half bus arrangement. The number of line positions and bus arrangement were selected to be indicative of a potential new substation layout. MISO will adjust its land requirements as required by specific potential project ideas.

New substation - land required (acre) (4 new line positions, breaker-and-a-half bus arrangement)							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Land required (acre)	1.50	1.75	2.00	2.50	4.00	8.00	20.00

For a scoping level cost estimates, the land required for both substation upgrades and new facilities will be reviewed for project specific considerations.

For the access road into a new substation, MISO uses Google Earth to estimate the length of the access road required, and the terrain type where the proposed substation site is located.

3.2.2 Equipment

Equipment costs are all the costs required to procure and install all the required equipment needed for substation upgrades or new substation facilities. For a substation upgrade, MISO assumes the quantity assumptions as shown in the tables below for a planning level cost estimation. For a scoping level cost estimate for a substation upgrade, MISO discusses all the equipment quantity assumptions with the existing substation owner:

Substation upgrade - equipment quantity assumptions (per line position)							
Estimate unit	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Circuit breaker	2	2	2	2	2	2	2
Disconnect switches	4	4	4	4	4	4	4
Voltage transformers (set of 3)	2	2	2	2	2	2	2
Bus support, bus and fittings (3-phase)	4	5	6	7	8	9	12
Deadend	1	1	1	1	1	1	1

Where the substation site is new, the equipment quantities required shown below are where the new sites has four new line positions, and a breaker-and-a-half bus arrangement. The number of line positions and bus arrangement were selected to be indicative of a potential new substation layout. MISO will adjust its equipment quantities as required by specific potential project ideas.

New substation - equipment quantity assumptions (4 new line positions, breaker-and-a-half bus arrangement)							
Estimate unit	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Circuit breaker	6	6	6	6	6	6	6
Disconnect switches	12	12	12	12	12	12	12
Voltage transformers (set of 3)	6	6	6	6	6	6	6
Bus support, bus and fittings (3-phase)	10	12	14	16	20	28	40
Deadend	4	4	4	4	4	4	4

For a scoping level cost estimate, the equipment quantities required for both substation upgrades and new facilities will be reviewed for project specific considerations.

For power transformers, reactors, capacitors and static VAR compensators, MISO considers them in the estimate as required for the potential project.

3.2.3 Protection and Control

Protection and control costs are all the costs required to procure and install the protection and control equipment for substation upgrades or new substation facilities. For a substation upgrade, MISO assumes the quantity assumptions as shown in the tables below for a planning level cost estimation. For a scoping level cost estimate for a substation upgrade, MISO discusses all the equipment quantity assumptions with the existing substation owner:

Substation upgrade – protection and control quantity assumptions (per line position)							
Estimate unit	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Control enclosure	0	0	0	0	0	0	0
Relay panels	2	2	2	2	2	2	2
Conduit (1000 feet)	1.0	1.25	1.50	1.75	2.0	2.25	2.50
Control cable (1000 feet)	10.0	12.5	15.0	17.5	20.0	22.5	25.0
Cable trench (1 foot)	100	125	150	175	200	225	250

Where the substation site is new, the protection and control quantities shown below are where the new sites has four new line positions, and a breaker-and-a-half bus arrangement. The number of line positions and bus arrangement were selected to be indicative of a potential new substation layout. MISO will adjust its protection and control quantities as required by specific potential project ideas.

New substation – protection and control quantity assumptions (4 line positions, breaker-and-a-half bus arrangement)							
Estimate unit	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Control enclosure	1	1	1	1	1	1	1
Relay panels	7	7	7	7	7	7	7
Conduit (1000 feet)	4	5	6	7	8	9	10
Control cable (1000 feet)	40	50	60	70	80	90	100
Cable trench (1 foot)	400	500	600	700	800	900	1000

For a scoping level cost estimate, the protection and control quantities required for both substation upgrades and new facilities will be reviewed for project specific considerations.

3.2.4 Overhead and professional services

Overhead and professional services encompass all the indirect overhead costs (e.g. computers, trucks) and all the professional services (e.g. surveying, drafting, engineering, project management, testing and commissioning) required for potential projects. This cost adder is to account for all the added costs that are spread out over the entire project and are not easily quantifiable by individual items.

3.3 Industry practices for project cost estimation

MISO researched industry practices for project cost estimating approaches, such as AACE (formerly the Association for the Advancement of Cost Engineering) International®, and referenced the following cost estimate classification system from AACE generic cost estimate classification matrix as a reference to MISO's provided project cost estimates. MISO's planning level estimates are considered equivalent to AACE Estimate Class 4. MISO's scoping level estimates are considered equivalent to AACE Estimate Class 3.

ESTIMATE CLASS	Primary Characteristic	Secondary Characteristic		
	MATURITY LEVEL OF PROJECT DEFINITION DELIVERABLES Expressed as % of complete definition	END USAGE Typical purpose of estimate	METHODOLOGY Typical estimating method	EXPECTED ACCURACY RANGE Typical variation in low and high ranges
Class 5	0% to 2%	Concept screening	Capacity factored, parametric models, judgment, or analogy	L: -20% to -50% H: +30% to +100%
Class 4	1% to 15%	Study or feasibility	Equipment factored or parametric models	L: -15% to -30% H: +20% to +50%
Class 3	10% to 40%	Budget authorization or control	Semi-detailed unit costs with assembly level line items	L: -10% to -20% H: +10% to +30%
Class 2	30% to 75%	Control or bid/tender	Detailed unit cost with forced detailed take-off	L: -5% to -15% H: +5% to +20%
Class 1	65% to 100%	Check estimate or bid/tender	Detailed unit cost with detailed take-off	L: -3% to -10% H: +3% to +15%

4. MISO's Cost Estimation Data

MISO's cost estimation data is as shown in section 4. The unit costs below do not include overhead and professional services, contingency, or AFUDC. The unit costs below are all in 2018 dollars.

4.1 Transmission Line Cost Data

4.1.1 Land and Right-of-Way

4.1.1.1 Land – Real Estate

Land required for right-of-way is calculated for new potential transmission line projects. MISO assumes that new land is required for all projects except transmission line rebuild projects. MISO has three categories of land costs: Pasture, Crop, and Urban/Suburban. Pasture land values are based on USDA published values. It is assumed that the Crop land is 3 times the unit cost of pasture land and the Urban/suburban land is 5 times the unit cost of pasture land. Based on its desktop analysis, MISO will determine the land type encountered for new potential transmission line projects. Land acquisition costs are considered separately in section 4.1.1.2.

State - land	Unit	Unit of measurement	Pasture land	Crop land	Urban/Suburban land
Arkansas	1	acre	\$2,583	\$7,749	\$12,915
Illinois	1	acre	\$3,583	\$10,749	\$17,915
Indiana	1	acre	\$2,655	\$7,965	\$13,275
Iowa	1	acre	\$3,178	\$9,534	\$15,890
Kentucky	1	acre	\$2,829	\$8,487	\$14,145
Louisiana	1	acre	\$2,768	\$8,304	\$13,840
Michigan	1	acre	\$2,614	\$7,842	\$13,070
Minnesota	1	acre	\$1,743	\$5,229	\$8,715
Mississippi	1	acre	\$2,265	\$6,795	\$11,325
Missouri	1	acre	\$2,019	\$6,057	\$10,095
Montana	1	acre	\$677	\$2,031	\$3,385
North Dakota	1	acre	\$871	\$2,613	\$4,355
South Dakota	1	acre	\$1,087	\$3,261	\$5,435
Texas	1	acre	\$1,691	\$5,073	\$8,455
Wisconsin	1	acre	\$2,409	\$7,227	\$12,045

4.1.1.2 Right-of-Way Acquisition

Real property rights for transmission projects include good faith negotiations with the land owners to acquire rights through easements, options, leases, fee purchases or eminent domain/condemnation. MISO assumes the cost of right-of-way acquisition to be \$9,225 per acre of land. The cost includes routing analysis, property tracts and mapping, land owner negotiations, land acquisition and condemnation fees. The land cost is considered separately in section 4.1.1.1.

4.1.1.3 Mobilization/Demobilization

This unit cost includes all the costs associated with mobilizing and demobilizing all equipment and people needed for construction of the project.

Single circuit and double circuit							
Unit detail	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Mobilization/Demobilization	\$100,000	\$100,000	\$150,000	\$150,000	\$200,000	\$250,000	\$300,000

4.1.1.4 Terrain

Terrain unit costs include all the costs associated with clearing the terrain along the route for the transmission line construction. The unit costs below are applied where portions of the potential project encounter the different terrain types listed below. These costs are the same for all voltage classes, and vary by the total amount of acreage. The three terrain categories are:

- Level ground with light vegetation: \$260/acre
- Forested: \$4,920/acre
- Wetland:
 - \$57,500/acre for matting including matting material procurement, placement and removal
 - \$46,125/acre for environmental mitigation credits
 - +20% increase to structure installation cost
 - +20% increase to structure foundation cost

4.1.1.5 Grading

Grading unit cost includes all associated costs for construction difficulties with access and workability where the grade of the land varies. MISO assumes the following costs for a given situation:

- Typical (0% to 50% slope): Included
- Mountainous (>50% slope): \$6,400/acre

4.1.1.6 Road Surfacing

Road surfacing unit cost includes all associated costs to provide access roads and dust control as needed for the transmission line. Where required, MISO assumes the cost for one mile of gravel road is as shown below:

- Road Surfacing: \$10,250/mile

4.1.2 Structures

4.1.2.1 Tangent Structures

Tangent structures are the most commonly used structures where the line alignment is relatively straight. Tangent structures support the conductor using a suspension insulator assembly. The suspension insulator assembly consists of insulator and hardware to provide necessary electrical insulation and strength for load transfer. The shieldwire (OPGW) is attached to the shieldwire suspension assembly. The tangent structures are designed for 0° to 2° line angle with the highest applicable NESC loading in the MISO region.

Material cost includes the cost of design, manufacture (material, labor, equipment) and delivery of the tangent structure and anchor bolts to site (laydown yard).

Installation cost is the cost to haul, assemble, and install the structure, insulator, line hardware and grounding assemblies. Inclusive of the installation costs is access to the structure location, and restoration.

Hardware/Insulators/Grounding cost includes material cost for insulator, line hardware and grounding assemblies.

Tangent structures are assumed to be supported on a concrete drilled pier foundation. Foundation size for a tangent structure is indicated as concrete volume required per structure in cubic yards. Foundation cost is the combination of the material cost and the installation cost for the foundation.

All cost components related to steel lattice tower and tubular pole tangent structures are described below. The unit costs are inclusive of all applicable taxes. The steel weights and foundation sizes MISO considers for its unit costs are an indicative value for structures at different voltage classes, and are not tied directly to any one structure type.

Single circuit tubular pole tangent structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	7,000	7,900	8,400	9,300	11,100	22,300	35,100
Foundation size	As shown	Cu. Yd	5.5	6.0	8.0	9.0	13.0	21.0	41.0

Single circuit tubular pole tangent structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$14,980	\$16,906	\$17,976	\$19,902	\$23,754	\$47,722	\$75,114
Installation	1	EA	\$22,470	\$25,359	\$26,964	\$29,853	\$35,631	\$71,583	\$112,671
Hardware/ Insulators/ Grounding	1	Set	\$3,930	\$4,585	\$4,913	\$5,568	\$6,550	\$8,764	\$9,594
Foundation	1	EA	\$7,040	\$7,680	\$10,240	\$11,520	\$16,640	\$26,880	\$52,480
Total cost	1	EA	\$48,420	\$54,530	\$60,093	\$66,843	\$82,575	\$154,949	\$249,859

Double circuit tubular pole tangent structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	11,300	12,700	13,500	14,900	18,600	36,000	50,300
Foundation size	As shown	Cu. Yd	8.0	10.0	14.5	17.5	23.0	46.5	78.5

Double circuit tubular pole tangent structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$24,182	\$27,178	\$28,890	\$31,886	\$39,804	\$77,040	\$107,642
Installation	1	EA	\$36,273	\$40,767	\$43,335	\$47,829	\$59,706	\$115,560	\$161,463
Hardware/ Insulators/ Grounding	1	Set	\$7,651	\$8,926	\$9,563	\$10,838	\$12,751	\$17,159	\$18,799
Foundation	1	EA	\$10,240	\$12,800	\$18,560	\$22,400	\$29,440	\$59,520	\$100,480
Total cost	1	EA	\$78,346	\$89,671	\$100,348	\$112,953	\$141,701	\$269,279	\$388,384

Single circuit lattice tower tangent structure design details									
Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	6,100	6,900	7,300	8,100	10,100	20,300	27,000
Foundation size	As shown	Cu. Yd	8.5	11.5	13.5	14.5	15.5	19.5	33.5

Single circuit lattice tower tangent structure unit costs									
Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$10,858	\$12,282	\$12,994	\$14,418	\$17,978	\$36,134	\$48,060
Installation	1	EA	\$16,287	\$18,423	\$19,491	\$21,627	\$26,967	\$54,201	\$72,090
Hardware/ Insulators/ Grounding	1	Set	\$3,930	\$4,585	\$4,913	\$5,568	\$6,550	\$8,764	\$9,594
Foundation	1	EA	\$10,880	\$14,720	\$17,280	\$18,560	\$19,840	\$24,960	\$42,880
Total cost	1	EA	\$41,955	\$50,010	\$54,678	\$60,173	\$71,335	\$124,059	\$172,624

Double circuit lattice tower tangent structure design details									
Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	9,200	10,400	11,000	12,200	15,200	36,000	41,900
Foundation size	As shown	Cu. Yd	13.0	17.0	19.5	21.0	22.0	31.5	48.5

Double circuit lattice tower tangent structure unit costs									
Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$16,376	\$18,512	\$19,580	\$21,716	\$27,056	\$64,080	\$74,582
Installation	1	EA	\$24,564	\$27,768	\$29,370	\$32,574	\$40,584	\$96,120	\$111,873
Hardware/ Insulators/ Grounding	1	Set	\$7,651	\$8,926	\$9,563	\$10,838	\$12,751	\$17,159	\$18,799
Foundation	1	EA	\$16,640	\$21,760	\$24,960	\$26,880	\$28,160	\$40,320	\$62,080
Total cost	1	EA	\$65,231	\$76,966	\$83,473	\$92,008	\$108,551	\$217,679	\$267,334

4.1.2.2 Small Angle Structures

Small angle structures are used where the line alignment changes direction and the line angle is between 2° and 15°. The structures are designed for the highest applicable NESC loading in the MISO region. Small angle structures support the conductor with suspension insulator assembly similar to tangent structures. The shieldwire (OPGW) is attached to the shieldwire suspension assembly near top of structure.

Material cost includes the cost of design, manufacture (material, labor, equipment) and delivery of the small angle structure and anchor bolts to site (laydown yard).

Installation cost is the cost to haul, assemble, and install the structure, insulator, line hardware and grounding assemblies. Inclusive of the installation costs is access to the structure location, and restoration.

Hardware/Insulators/Grounding cost includes material cost for insulators, line hardware and grounding assemblies.

Small angle structures are assumed to be supported on a concrete drilled pier foundation. Foundation size for a Small angle structure is measured as concrete volume required per structure in cubic yards. Foundation cost is the combination of the material cost and the installation cost for the foundation.

All cost components related to steel lattice tower and tubular pole Small angle structures are described below. The unit costs are inclusive of all applicable taxes. The steel weights and foundation sizes MISO considers for its unit costs are an indicative value for structures at different voltage classes, and are not tied directly to any one structure type.

Single circuit tubular steel pole small angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	8,400	9,500	10,100	11,200	13,300	31,800	45,600
Foundation size	As shown	Cu. Yd	6.5	7.5	9.5	10.5	15.0	24.0	45.5

Single circuit tubular steel pole small angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$17,976	\$20,330	\$21,614	\$23,968	\$28,462	\$68,052	\$97,584
Installation	1	EA	\$26,964	\$30,495	\$32,421	\$35,952	\$42,693	\$102,078	\$146,376
Hardware/ Insulators/ Grounding	1	Set	\$3,930	\$4,585	\$4,913	\$5,568	\$6,550	\$8,764	\$9,594
Foundation	1	EA	\$8,320	\$9,600	\$12,160	\$13,440	\$19,200	\$30,720	\$58,240
Total cost	1	EA	\$57,190	\$65,010	\$71,108	\$78,928	\$96,905	\$209,614	\$311,794

Double circuit tubular steel pole Small Angle Structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	12,400	14,000	14,900	16,400	20,500	39,600	55,300
Foundation size	As shown	Cu. Yd	9.5	12.0	17.0	20.0	26.0	51.5	85.5

Double circuit tubular steel pole small angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$26,536	\$29,960	\$31,886	\$35,096	\$43,870	\$84,744	\$118,342
Installation	1	EA	\$39,804	\$44,940	\$47,829	\$52,644	\$65,805	\$127,116	\$177,513
Hardware/ Insulators/ Grounding	1	Set	\$7,651	\$8,926	\$9,563	\$10,838	\$12,751	\$17,159	\$18,799
Foundation	1	EA	\$12,160	\$15,360	\$21,760	\$25,600	\$33,280	\$65,920	\$109,440
Total cost	1	EA	\$86,151	\$99,186	\$111,038	\$124,178	\$155,706	\$294,939	\$424,094

Single circuit lattice tower small angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	6,800	7,700	8,100	9,000	11,300	22,600	30,100
Foundation size	As shown	Cu. Yd	12.5	14.5	17.0	18.5	19.5	24.0	47.0

Single circuit lattice tower small angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$12,104	\$13,706	\$14,418	\$16,020	\$20,114	\$40,228	\$53,578
Installation	1	EA	\$18,156	\$20,559	\$21,627	\$24,030	\$30,171	\$60,342	\$80,367
Hardware/ Insulators/ Grounding	1	Set	\$3,930	\$4,585	\$4,913	\$5,568	\$6,550	\$8,764	\$9,594
Foundation	1	EA	\$16,000	\$18,560	\$21,760	\$23,680	\$24,960	\$30,720	\$60,160
Total cost	1	EA	\$50,190	\$57,410	\$62,718	\$69,298	\$81,795	\$140,054	\$203,699

Double circuit lattice tower small angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	10,600	12,000	12,700	14,000	17,500	40,500	47,100
Foundation size	As shown	Cu. Yd	18.0	22.0	27.0	28.5	30.0	39.5	59.0

Double circuit lattice tower small angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$18,868	\$21,360	\$22,606	\$24,920	\$31,150	\$72,090	\$83,838
Installation	1	EA	\$28,302	\$32,040	\$33,909	\$37,380	\$46,725	\$108,135	\$125,757
Hardware/ Insulators/ Grounding	1	Set	\$7,651	\$8,926	\$9,563	\$10,838	\$12,751	\$17,159	\$18,799
Foundation	1	EA	\$23,040	\$28,160	\$34,560	\$36,480	\$38,400	\$50,560	\$75,520
Total cost	1	EA	\$77,861	\$90,486	\$100,638	\$109,618	\$129,026	\$247,944	\$303,914

4.1.2.3 Large Angle Structures

Large angle structures are used where the line alignment changes direction and the line angle is between 15° and 45°. The structures are designed for the highest applicable NESC loading in the MISO region. Large angle structures support the conductor with a suspension insulator assembly similar to tangent and small angle structures. The shieldwire (OPGW) is attached to a shieldwire suspension assembly near top of structure.

Material cost includes the cost of design, manufacture (material, labor, equipment) and delivery of the large angle structure and anchor bolts to site (laydown yard).

Installation cost is the cost to haul, assemble, and install the structure, insulator, line hardware and grounding assemblies. Inclusive of the installation costs is access to the structure location, and restoration.

Hardware/Insulators/Grounding cost includes material cost for insulators, line hardware and grounding assemblies.

A large angle structure is assumed to be supported on a concrete drilled pier foundation. Foundation size for a large angle structure is measured in concrete volume required per structure in cubic yards. Foundation cost is the combination of the material cost and the installation cost for the foundation.

All cost components related to steel lattice tower and tubular pole large angle structures are described below. The unit costs are inclusive of all applicable taxes. The steel weights and foundation sizes MISO considers for its unit costs are an indicative value for structures at different voltage classes, and are not tied directly to any one structure type.

Single circuit tubular steel pole large angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	11,600	13,000	13,900	15,300	18,300	37,900	59,700
Foundation size	As shown	Cu. Yd	9.0	10.5	13.0	14.0	19.5	30.0	54.5

Single circuit tubular steel pole large angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$24,824	\$27,820	\$29,746	\$32,742	\$39,162	\$81,106	\$127,758
Installation	1	EA	\$37,236	\$41,730	\$44,619	\$49,113	\$58,743	\$121,659	\$191,637
Hardware/ Insulators/ Grounding	1	Set	\$3,930	\$4,585	\$4,913	\$5,568	\$6,550	\$8,764	\$9,594
Foundation	1	EA	\$11,520	\$13,440	\$16,640	\$17,920	\$24,960	\$38,400	\$69,760
Total cost	1	EA	\$77,510	\$87,575	\$95,918	\$105,343	\$129,415	\$249,929	\$398,749

Double circuit tubular steel pole large angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	15,000	16,800	17,900	19,700	24,600	47,700	70,400
Foundation size	As shown	Cu. Yd	13.0	15.5	21.5	25.5	32.5	61.0	99.0

Double circuit tubular steel pole large angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$32,100	\$35,952	\$38,306	\$42,158	\$52,644	\$102,078	\$150,656
Installation	1	EA	\$48,150	\$53,928	\$57,459	\$63,237	\$78,966	\$153,117	\$225,984
Hardware/ Insulators/ Grounding	1	Set	\$7,651	\$8,926	\$9,563	\$10,838	\$12,751	\$17,159	\$18,799
Foundation	1	EA	\$16,640	\$19,840	\$27,520	\$32,640	\$41,600	\$78,080	\$126,720
Total cost	1	EA	\$104,541	\$118,646	\$132,848	\$148,873	\$185,961	\$350,434	\$522,159

Single circuit lattice tower large angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	9,200	10,400	11,000	12,200	15,200	30,500	39,800
Foundation size	As shown	Cu. Yd	16.0	19.0	19.5	22.0	24.5	39.0	72.5

Single circuit lattice tower large angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$16,376	\$18,512	\$19,580	\$21,716	\$27,056	\$54,290	\$70,844
Installation	1	EA	\$24,564	\$27,768	\$29,370	\$32,574	\$40,584	\$81,435	\$106,266
Hardware/ Insulators/ Grounding	1	Set	\$3,930	\$4,585	\$4,913	\$5,568	\$6,550	\$8,764	\$9,594
Foundation	1	EA	\$20,480	\$24,320	\$24,960	\$28,160	\$31,360	\$49,920	\$92,800
Total cost	1	EA	\$65,350	\$75,185	\$78,823	\$88,018	\$105,550	\$194,409	\$279,504

Double circuit lattice tower large angle structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	13,800	15,600	16,500	18,300	22,800	53,100	62,900
Foundation size	As shown	Cu. Yd	22.5	28.0	34.5	37.5	46.5	59.0	87.5

Double circuit lattice tower large angle structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$24,564	\$27,768	\$29,370	\$32,574	\$40,584	\$94,518	\$111,962
Installation	1	EA	\$36,846	\$41,652	\$44,055	\$48,861	\$60,876	\$141,777	\$167,943
Hardware/ Insulators/ Grounding	1	Set	\$7,651	\$8,926	\$9,563	\$10,838	\$12,751	\$17,159	\$18,799
Foundation	1	EA	\$28,800	\$35,840	\$44,160	\$48,000	\$59,520	\$75,520	\$112,000
Total cost	1	EA	\$97,861	\$114,186	\$127,148	\$140,273	\$173,731	\$328,974	\$410,704

4.1.2.4 Strain Structures

Strain structures are partial deadend structures and not designed for full terminal loads. They are designed to withstand some unbalanced wire tensions in one direction of one or all wires on one face of the structure. The strain structures are designed for line angle between 5° to 45°. The structures are designed for the highest applicable NESC loading in the MISO region.

Material cost includes the cost of design, manufacture (material, labor, equipment) and delivery of the strain structure and anchor bolts to site (laydown yard).

Installation cost is the cost to haul, assemble, and install the structure, insulator, line hardware and grounding assemblies. Inclusive of the installation costs is access to the structure location, and restoration.

Hardware/Insulators/Grounding cost includes material cost for insulators, line hardware and grounding assemblies.

Strain structures are assumed to be supported on a concrete drilled pier foundation. Foundation size for a strain structure is measured in concrete volume required per structure in cubic yards. Foundation cost is the combination of the material cost and the installation cost for the foundation.

All cost components related to steel lattice tower and tubular pole strain structures are described below. The unit costs are inclusive of all applicable taxes. The steel weights and foundation sizes MISO considers for its unit costs are an indicative value for structures at different voltage classes, and are not tied directly to any one structure type.

Single circuit tubular steel pole strain structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	14,000	15,800	16,800	18,600	22,200	42,400	66,700
Foundation size	As shown	Cu. Yd	11.0	12.0	15.0	16.5	22.5	33.5	60.0

Single circuit tubular steel pole strain structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$29,960	\$33,812	\$35,952	\$39,804	\$47,508	\$90,736	\$142,738
Installation	1	EA	\$44,940	\$50,718	\$53,928	\$59,706	\$71,262	\$136,104	\$214,107
Hardware/ Insulators/ Grounding	1	Set	\$7,749	\$9,041	\$9,686	\$10,978	\$12,915	\$31,498	\$49,549
Foundation	1	EA	\$14,080	\$15,360	\$19,200	\$21,120	\$28,800	\$42,880	\$76,800
Total cost	1	EA	\$96,729	\$108,931	\$118,766	\$131,608	\$160,485	\$301,218	\$483,194

Double circuit tubular steel pole strain structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	16,700	18,700	19,900	22,000	27,400	54,000	75,500
Foundation size	As shown	Cu. Yd	15.5	18.5	25.0	29.5	37.0	68.5	109.0

Double circuit tubular steel pole strain structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$35,738	\$40,018	\$42,586	\$47,080	\$58,636	\$115,560	\$161,570
Installation	1	EA	\$53,607	\$60,027	\$63,879	\$70,620	\$87,954	\$173,340	\$242,355
Hardware/ Insulators/ Grounding	1	Set	\$15,283	\$17,830	\$19,104	\$21,651	\$25,471	\$62,648	\$98,738
Foundation	1	EA	\$19,840	\$23,680	\$32,000	\$37,760	\$47,360	\$87,680	\$139,520
Total cost	1	EA	\$124,468	\$141,555	\$157,569	\$177,111	\$219,421	\$439,228	\$642,183

Single circuit lattice tower strain structure design details									
Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	10,400	11,700	12,400	13,800	17,200	34,500	45,900
Foundation size	As shown	Cu. Yd	21.5	25.0	25.5	28.5	34.0	48.5	96.0

Single circuit lattice tower strain structure unit costs									
Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$18,512	\$20,826	\$22,072	\$24,564	\$30,616	\$61,410	\$81,702
Installation	1	EA	\$27,768	\$31,239	\$33,108	\$36,846	\$45,924	\$92,115	\$122,553
Hardware/ Insulators/ Grounding	1	Set	\$7,749	\$9,041	\$9,686	\$10,978	\$12,915	\$31,498	\$49,549
Foundation	1	EA	\$27,520	\$32,000	\$32,640	\$36,480	\$43,520	\$62,080	\$122,880
Total cost	1	EA	\$81,549	\$93,106	\$97,506	\$108,868	\$132,975	\$247,103	\$376,684

Double circuit lattice tower strain structure design details									
Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	16,100	18,200	19,300	21,400	26,600	61,200	71,200
Foundation size	As shown	Cu. Yd	28.5	34.5	43.0	48.5	70.5	86.5	126.5

Double circuit lattice tower strain structure unit costs									
Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$28,658	\$32,396	\$34,354	\$38,092	\$47,348	\$108,936	\$126,736
Installation	1	EA	\$42,987	\$48,594	\$51,531	\$57,138	\$71,022	\$163,404	\$190,104
Hardware/ Insulators/ Grounding	1	Set	\$15,283	\$17,830	\$19,104	\$21,651	\$25,471	\$62,648	\$98,738
Foundation	1	EA	\$36,480	\$44,160	\$55,040	\$62,080	\$90,240	\$110,720	\$161,920
Total cost	1	EA	\$123,408	\$142,980	\$160,029	\$178,961	\$234,081	\$445,708	\$577,498

4.1.2.5 Deadend Structures

Deadend structures are designed for full terminal loads for all wires and line angles between 0° and 90°. The structures are designed for the highest applicable NESC loading in the MISO region.

Material cost includes the cost to design, manufacture (material, labor, equipment) and delivery of the deadend structure and anchor bolts to site (laydown yard).

Installation cost is the cost to haul, assemble, and install the structure, insulator, line hardware and grounding assemblies. Inclusive of the installation costs is access to the structure location, and restoration.

Hardware/Insulators/Grounding cost includes material cost for insulator, line hardware and grounding assemblies.

Deadend structures are assumed to be supported on a concrete drilled pier foundation. Foundation size for a deadend structure is measured in concrete volume required per structure in cubic yards. Foundation cost is the combination of the material cost and the installation cost for the foundation.

All cost components related to steel lattice towers and tubular pole deadend structures are described below. The unit costs are inclusive of all applicable taxes. The steel weights and foundation sizes MISO considers for its unit costs are an indicative value for structures at different voltage classes, and are not tied directly to any one structure type.

Single circuit tubular steel pole deadend structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	20,400	23,000	24,500	27,100	32,400	48,100	80,700
Foundation size	As shown	Cu. Yd	15.0	16.5	20.0	21.5	29.0	41.5	72.0

Single circuit tubular steel pole deadend structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$43,656	\$49,220	\$52,430	\$57,994	\$69,336	\$102,934	\$172,698
Installation	1	EA	\$65,484	\$73,830	\$78,645	\$86,991	\$104,004	\$154,401	\$259,047
Hardware/ Insulators/ Grounding	1	Set	\$7,749	\$9,041	\$9,686	\$10,978	\$12,915	\$31,498	\$49,549
Foundation	1	EA	\$19,200	\$21,120	\$25,600	\$27,520	\$37,120	\$53,120	\$92,160
Total cost	1	EA	\$136,089	\$153,211	\$166,361	\$183,483	\$223,375	\$341,953	\$573,454

Double circuit tubular steel pole deadend structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	26,000	29,200	31,100	34,300	42,800	84,600	118,200
Foundation size	As shown	Cu. Yd	20.0	24.0	32.0	37.0	46.0	81.5	127.0

Double circuit tubular steel pole strain structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$55,640	\$62,488	\$66,554	\$73,402	\$91,592	\$181,044	\$252,948
Installation	1	EA	\$83,460	\$93,732	\$99,831	\$110,103	\$137,388	\$271,566	\$379,422
Hardware/ Insulators/ Grounding	1	Set	\$15,283	\$17,830	\$19,104	\$21,651	\$25,471	\$62,648	\$98,738
Foundation	1	EA	\$25,600	\$30,720	\$40,960	\$47,360	\$58,880	\$104,320	\$162,560
Total cost	1	EA	\$179,983	\$204,770	\$226,449	\$252,516	\$313,331	\$619,578	\$893,668

Single circuit lattice tower deadend structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	13,400	15,200	16,100	17,800	22,200	44,700	59,400
Foundation size	As shown	Cu. Yd	33.5	38.0	39.0	43.0	52.0	90.0	176.0

Single circuit lattice tower deadend structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$23,852	\$27,056	\$28,658	\$31,684	\$39,516	\$79,566	\$105,732
Installation	1	EA	\$35,778	\$40,584	\$42,987	\$47,526	\$59,274	\$119,349	\$158,598
Hardware/ Insulators/ Grounding	1	Set	\$7,749	\$9,041	\$9,686	\$10,978	\$12,915	\$31,498	\$49,549
Foundation	1	EA	\$42,880	\$48,640	\$49,920	\$55,040	\$66,560	\$115,200	\$225,280
Total cost	1	EA	\$110,259	\$125,321	\$131,251	\$145,228	\$178,265	\$345,613	\$539,159

Double circuit lattice tower deadend structure design details

Design detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Steel weight	As shown	Pounds	21,200	23,900	25,300	28,100	35,000	79,200	92,200
Foundation size	As shown	Cu. Yd	43.0	50.5	61.5	68.5	99.0	125.0	236.0

Double circuit lattice tower strain structure unit costs

Unit detail	Unit	Unit of measurement	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Material	1	EA	\$37,736	\$42,542	\$45,034	\$50,018	\$62,300	\$140,976	\$164,116
Installation	1	EA	\$56,604	\$63,813	\$67,551	\$75,027	\$93,450	\$211,464	\$246,174
Hardware/ Insulators/ Grounding	1	Set	\$15,283	\$17,830	\$19,104	\$21,651	\$25,471	\$62,648	\$98,738
Foundation	1	EA	\$55,040	\$64,640	\$78,720	\$87,680	\$126,720	\$160,000	\$302,080
Total cost	1	EA	\$164,663	\$188,825	\$210,409	\$234,376	\$307,941	\$575,088	\$811,108

4.1.3 Conductor, Shieldwire, and Accessories

4.1.3.1 Conductor

Conductors in transmission lines constitute a considerable portion of the overall material and labor costs of the line. MISO considers ACSR (Aluminum Conductor Steel Reinforce) and ACSS (Aluminum Conductor Steel Supported) to account for conductor costs. Where required, MISO would consider the cost for T2 to be equivalent to two conductors of that size to the same cost when creating its cost estimate.

Material cost is the cost of manufacturing and deliver conductor to site (laydown yard).

Installation cost is the cost to haul conductor reels, install, and sag and clip conductor on transmission structures.

The unit costs are inclusive of all applicable taxes.

Conductor	Unit	Unit of measurement	Conductor type	Installation	Material	Total cost
266.8 "Waxwing" per phase	1000	feet	ACSR	\$646	\$526	\$1,172
266.8 "Partridge" per phase	1000	feet	ACSR	\$820	\$634	\$1,454
336.4 "Merlin" per phase	1000	feet	ACSR	\$789	\$561	\$1,350
336.4 "Linnet" per phase	1000	feet	ACSR	\$964	\$646	\$1,609
336.4 "Oriole" per phase	1000	feet	ACSR	\$1,035	\$806	\$1,841
397.5 "Chickadee" per phase	1000	feet	ACSR	\$902	\$692	\$1,594
397.5 "Ibis" per phase	1000	feet	ACSR	\$1,097	\$831	\$1,928
397.5 "Lark" per phase	1000	feet	ACSR	\$1,230	\$820	\$2,050
477 "Pelican" per phase	1000	feet	ACSR	\$1,107	\$811	\$1,918
477 "Flicker" per phase	1000	feet	ACSR	\$1,158	\$779	\$1,937
477 "Hawk" per phase	1000	feet	ACSR	\$1,281	\$969	\$2,250
477 "Hen" per phase	1000	feet	ACSR	\$1,394	\$1,079	\$2,473
556.5 "Osprey" per phase	1000	feet	ACSR	\$1,240	\$974	\$2,214
556.5 "Parakeet" per phase	1000	feet	ACSR	\$1,425	\$1,143	\$2,568
556.5 "Dove" per phase	1000	feet	ACSR	\$1,486	\$1,080	\$2,567
636 "Kingbird" per phase	1000	feet	ACSR	\$1,394	\$941	\$2,335
636 "Rook" per phase	1000	feet	ACSR	\$1,609	\$1,066	\$2,675
636 "Grosbeak" per phase	1000	feet	ACSR	\$1,671	\$1,222	\$2,893
666.6 "Flamingo" per phase	1000	feet	ACSR	\$1,845	\$1,230	\$3,075
795 "Coot" per phase	1000	feet	ACSR	\$1,722	\$1,246	\$2,968
795 "Tern" per phase	1000	feet	ACSR	\$1,753	\$1,179	\$2,932
795 "Cuckoo" per phase	1000	feet	ACSR	\$1,978	\$1,312	\$3,290
795 "Condor" per phase	1000	feet	ACSR	\$2,060	\$1,363	\$3,424
795 "Drake" per phase	1000	feet	ACSR	\$2,040	\$1,476	\$3,516
900 "Canary" per phase	1000	feet	ACSR	\$2,501	\$1,671	\$4,172
954 "Rail" per phase	1000	feet	ACSR	\$2,060	\$1,557	\$3,617
954 "Cardinal" per phase	1000	feet	ACSR	\$2,624	\$1,706	\$4,330
1033.5 "Ortolan" per phase	1000	feet	ACSR	\$2,224	\$1,708	\$3,932
1033.5 "Curlew" per phase	1000	feet	ACSR	\$2,819	\$1,883	\$4,702
1113 "Bluejay" per phase	1000	feet	ACSR	\$2,378	\$1,814	\$4,192
1192.5 "Bunting" per phase	1000	feet	ACSR	\$2,542	\$1,691	\$4,233
1272 "Bittern" per phase	1000	feet	ACSR	\$2,706	\$1,960	\$4,666
1272 "Pheasant" per phase	1000	feet	ACSR	\$3,219	\$2,142	\$5,361
1351.5 "Dipper" per phase	1000	feet	ACSR	\$2,850	\$2,120	\$4,969
1351.5 "Martin" per phase	1000	feet	ACSR	\$3,342	\$2,627	\$5,969
1431 "Bobolink" per phase	1000	feet	ACSR	\$3,280	\$2,403	\$5,683
1590 "Lapwing" per phase	1000	feet	ACSR	\$3,659	\$2,478	\$6,138
1590 "Falcon" per phase	1000	feet	ACSR	\$4,264	\$2,925	\$7,189
1780 "Chukar" per phase	1000	feet	ACSR	\$5,217	\$3,188	\$8,405
2156 "Bluebird" per phase	1000	feet	ACSR	\$6,222	\$3,755	\$9,976
2167 "Kiwi" per phase	1000	feet	ACSR	\$5,566	\$3,400	\$8,966
2312 "Thrasher" per phase	1000	feet	ACSR	\$5,863	\$3,895	\$9,758
2515 "Joree" per phase	1000	feet	ACSR	\$6,201	\$4,140	\$10,341

Conductor	Unit	Unit of measurement	Conductor type	Installation	Material	Total cost
266.8 "Waxwing" per phase	1000	feet	ACSS	\$646	\$513	\$1,158
266.8 "Partridge" per phase	1000	feet	ACSS	\$820	\$656	\$1,476
336.4 "Merlin" per phase	1000	feet	ACSS	\$789	\$625	\$1,415
336.4 "Linnet" per phase	1000	feet	ACSS	\$964	\$748	\$1,712
336.4 "Oriole" per phase	1000	feet	ACSS	\$1,035	\$830	\$1,866
397.5 "Chickadee" per phase	1000	feet	ACSS	\$902	\$728	\$1,630
397.5 "Ibis" per phase	1000	feet	ACSS	\$1,097	\$887	\$1,983
397.5 "Lark" per phase	1000	feet	ACSS	\$1,230	\$984	\$2,214
477 "Pelican" per phase	1000	feet	ACSS	\$1,107	\$892	\$1,999
477 "Flicker" per phase	1000	feet	ACSS	\$1,158	\$933	\$2,091
477 "Hawk" per phase	1000	feet	ACSS	\$1,281	\$1,035	\$2,317
477 "Hen" per phase	1000	feet	ACSS	\$1,394	\$1,107	\$2,501
556.5 "Osprey" per phase	1000	feet	ACSS	\$1,240	\$984	\$2,224
556.5 "Parakeet" per phase	1000	feet	ACSS	\$1,425	\$1,138	\$2,563
556.5 "Dove" per phase	1000	feet	ACSS	\$1,486	\$1,189	\$2,675
636 "Kingbird" per phase	1000	feet	ACSS	\$1,394	\$1,107	\$2,501
636 "Rook" per phase	1000	feet	ACSS	\$1,609	\$1,281	\$2,891
636 "Grosbeak" per phase	1000	feet	ACSS	\$1,671	\$1,333	\$3,003
666.6 "Flamingo" per phase	1000	feet	ACSS	\$1,845	\$1,476	\$3,321
795 "Coot" per phase	1000	feet	ACSS	\$1,722	\$1,384	\$3,106
795 "Tern" per phase	1000	feet	ACSS	\$1,753	\$1,404	\$3,157
795 "Cuckoo" per phase	1000	feet	ACSS	\$1,978	\$1,579	\$3,557
795 "Condor" per phase	1000	feet	ACSS	\$1,978	\$1,579	\$3,557
795 "Drake" per phase	1000	feet	ACSS	\$2,040	\$1,485	\$3,525
900 "Canary" per phase	1000	feet	ACSS	\$2,501	\$1,630	\$4,131
954 "Rail" per phase	1000	feet	ACSS	\$2,060	\$1,585	\$3,645
954 "Cardinal" per phase	1000	feet	ACSS	\$2,624	\$1,757	\$4,381
1033.5 "Ortolan" per phase	1000	feet	ACSS	\$2,224	\$2,112	\$4,336
1033.5 "Curlew" per phase	1000	feet	ACSS	\$2,819	\$1,784	\$4,602
1113 "Bluejay" per phase	1000	feet	ACSS	\$2,378	\$2,265	\$4,643
1192.5 "Bunting" per phase	1000	feet	ACSS	\$2,542	\$1,896	\$4,438
1272 "Bittern" per phase	1000	feet	ACSS	\$2,706	\$2,030	\$4,736
1272 "Pheasant" per phase	1000	feet	ACSS	\$3,219	\$2,346	\$5,565
1351.5 "Dipper" per phase	1000	feet	ACSS	\$2,850	\$2,573	\$5,422
1351.5 "Martin" per phase	1000	feet	ACSS	\$3,342	\$2,286	\$5,627
1431 "Bobolink" per phase	1000	feet	ACSS	\$3,280	\$2,675	\$5,955
1590 "Lapwing" per phase	1000	feet	ACSS	\$3,659	\$2,624	\$6,283
1590 "Falcon" per phase	1000	feet	ACSS	\$4,264	\$2,927	\$7,191
1780 "Chukar" per phase	1000	feet	ACSS	\$5,217	\$3,413	\$8,631
2156 "Bluebird" per phase	1000	feet	ACSS	\$6,222	\$4,172	\$10,394
2167 "Kiwi" per phase	1000	feet	ACSS	\$5,566	\$4,971	\$10,537
2312 "Thrasher" per phase	1000	feet	ACSS	\$5,863	\$4,459	\$10,322
2515 "Joree" per phase	1000	feet	ACSS	\$6,201	\$4,674	\$10,875

4.1.3.2 OPGW

Optical Groundwire (OPGW) is installed at the top of structures to protect the conductors below from direct lightning strikes, and includes fiber optic cable. The material cost is the cost of manufacturing and delivery of OPGW to site (laydown yard). The installation cost is the cost to haul OPGW reels, install, and sag and clip conductor on transmission structures.

The unit cost is inclusive of all applicable taxes.

Wire	Unit	Unit of measurement	Installation	Material	Total cost
OPGW	1000	feet	\$3,475	\$2,317	\$5,791

4.1.3.1 Shieldwire

Shieldwire is installed at the top of structures to protect the conductors below from direct lightning strikes. The material cost is the cost of manufacturing and delivery of shieldwire to site (laydown yard). The installation cost is the cost to haul shieldwire reels, install, and sag and clip conductor on transmission structures.

The unit cost is inclusive of all applicable taxes.

Wire	Unit	Unit of measurement	Installation	Material	Total cost
Shieldwire	1000	feet	\$769	\$513	\$1,281

4.1.3.1 Accessories

Accessories are the sleeves, spacers, and dampers material and installation cost required for a transmission line. The unit cost per conductor length is as below:

- Accessories: \$1,200/mile

4.1.4 Overhead and professional services

4.1.4.1 Overhead and professional services

Overhead and professional services encompass all the indirect overhead costs (e.g. computers, trucks) and all the professional services (e.g. surveying, drafting, engineering, project management, testing and commissioning) required for potential projects. This cost adder is to account for all the added costs that are spread out over the entire project and are not easily quantifiable by individual items. MISO aggregates all the overhead and professionals services into one unit cost in order to facilitate discussion on the larger cost adder value that MISO considers for these costs. The contributions of these services are at a cost of 10% of the construction costs for the project.

4.2 Substation Cost Data

4.2.1 Site Work

4.2.1.1 Land – Real Estate

Land required for substation upgrades and new substations are calculated based on a given voltage level and quantity of equipment required. MISO has three categories of land costs: Pasture; Crop; Urban/Suburban. Pasture land values are based on USDA published values. It is assumed that the Crop land is 3 times the unit cost of pasture land and the Urban/suburban land is 5 times the unit cost of pasture land. Based on its desktop analysis, MISO will determine the land type encountered for new potential transmission line projects. Land acquisition costs are considered separately in section 4.2.1.5.

State - land	Unit	Unit of measurement	Pasture land	Crop land	Urban/Suburban land
Arkansas	1	acre	\$2,583	\$7,749	\$12,915
Illinois	1	acre	\$3,583	\$10,749	\$17,915
Indiana	1	acre	\$2,655	\$7,965	\$13,275
Iowa	1	acre	\$3,178	\$9,534	\$15,890
Kentucky	1	acre	\$2,829	\$8,487	\$14,145
Louisiana	1	acre	\$2,768	\$8,304	\$13,840
Michigan	1	acre	\$2,614	\$7,842	\$13,070
Minnesota	1	acre	\$1,743	\$5,229	\$8,715
Mississippi	1	acre	\$2,265	\$6,795	\$11,325
Missouri	1	acre	\$2,019	\$6,057	\$10,095
Montana	1	acre	\$677	\$2,031	\$3,385
North Dakota	1	acre	\$871	\$2,613	\$4,355
South Dakota	1	acre	\$1,087	\$3,261	\$5,435
Texas	1	acre	\$1,691	\$5,073	\$8,455
Wisconsin	1	acre	\$2,409	\$7,227	\$12,045

4.2.1.2 Mobilization/Demobilization

This unit cost includes all the costs associated with mobilizing and demobilizing all equipment and people needed for construction of the project.

Mobilization/Demobilization									
Unit details	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
New site	1	EA	\$262,660	\$262,660	\$262,660	\$262,660	\$262,660	\$262,660	\$262,660
Existing site upgrade	1	EA	\$157,590	\$157,590	\$157,590	\$157,590	\$157,590	\$157,590	\$157,590

4.2.1.3 Clearing and Grubbing

Clearing and grubbing unit cost is the cost to clear all the vegetation off the land for use. MISO uses Google Earth to estimate the land type encountered. The three clearing and grubbing categories are:

- Level ground with light vegetation: \$260/acre
- Forested: \$4,920/acre
- Wetland:
 - \$57,500/acre for matting including matting material procurement, placement and removal
 - \$46,200/acre for environmental mitigation credits

4.2.1.4 Grading

Grading is the unit cost to prepare the site to be constructed upon. It is assumed that there is balanced cut and fill, and no import or export of soil required.

- Grading: \$75,500/acre

4.2.1.5 Site/Fence/Grounding

Site/Fence/Grounding is the unit cost that includes all the general site work (including land selection and acquisition, lightning protection, and station service power), the security fence around the exterior of the yard, and the grounding system (inclusive of ground grid and top rock). Where more site components are required (e.g. security cameras, specialized gates, access protection) MISO will add those costs to its cost estimate.

- Site/Fence/Grounding: \$280,000/acre

4.2.1.6 Access Roads

Access Roads allow entry to the substation site from the nearest drivable public road. The unit price for access roads depends on the terrain type of where the substation is constructed and is as shown below.

- Level ground with light vegetation: \$10,250/mile
- Forested: \$51,250/mile
- Wetland: \$512,500/mile

4.2.2 Equipment

4.2.2.1 Circuit Breaker(s)

This unit cost is for the cost associated with one circuit breaker. Material cost is the cost to procure and deliver one circuit breaker to site (laydown yard). Installation cost includes: assembly; placement on the foundation; connection to the main ground grid; above ground conduit to connect the below grade conduit to the equipment control box; wiring to land the control cables on the terminal blocks in the equipment control box; jumpers and fittings to connect to substation bus. Foundation size is the amount of cubic yards of concrete required for the foundation. Foundation cost is the combination of the material and installation cost for the foundation. The unit costs are inclusive of all applicable taxes.

Circuit Breaker (3-phase) design details									
Design detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Foundation size	As shown	Cu. Yd	3.6	4.5	5.3	6.7	8.0	8.8	19.8

Circuit Breaker (3-phase) unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	EA	\$50,840	\$63,550	\$72,693	\$82,000	\$100,963	\$306,906	\$403,830
Installation	1	EA	\$10,168	\$12,710	\$14,539	\$16,400	\$20,193	\$61,381	\$80,766
Foundation	1	EA	\$4,608	\$5,760	\$6,784	\$8,576	\$10,240	\$11,264	\$25,344
Total cost	1	EA	\$65,616	\$82,020	\$94,016	\$106,976	\$131,395	\$379,551	\$509,939

4.2.2.2 Disconnect Switch(es)

This unit cost is the cost associated with one disconnect switch. Material cost is the cost to procure and deliver one disconnect switch to site (laydown yard). Installation cost includes: assembly; placement on the steel stand; and connection to main ground grid; jumpers and fittings to connect to substation bus. Steel stand material cost is the cost to procure and deliver one steel stand for a disconnect switch to site (laydown yard). Steel stand installation cost is the cost to place the steel stand on the foundation. Foundation size is the amount of cubic yards of concrete required for the foundation. Foundation cost is the combination of the material and installation cost for the foundation. The unit costs are inclusive of all applicable taxes.

Disconnect switch (3-phase) design details									
Design detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Foundation size	As shown	Cu. Yd	3.4	4.2	5.2	6.5	7.8	8.0	18.0

Disconnect switch (3-phase) unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	EA	\$19,147	\$23,934	\$28,270	\$32,308	\$36,347	\$51,691	\$67,845
Installation	1	EA	\$8,616	\$10,770	\$12,721	\$14,539	\$16,356	\$23,261	\$30,530
Steel stand material	1	EA	\$4,510	\$5,638	\$6,458	\$7,267	\$8,077	\$16,154	\$22,612
Steel stand installation	1	EA	\$5,187	\$6,483	\$7,426	\$8,357	\$9,289	\$18,577	\$26,003
Foundation	1	EA	\$4,352	\$5,376	\$6,656	\$8,320	\$9,984	\$10,240	\$23,040
Total cost	1	EA	\$41,812	\$52,201	\$61,530	\$70,791	\$80,052	\$119,923	\$170,030

4.2.2.3 Voltage Transformer(s)

This unit cost is the cost associated with one set of three voltage transformers. Material cost is the cost to procure and deliver one set of three voltage transformers to site (laydown yard). Installation cost includes: assembly; placement on the steel stand; connection to the main ground grid; above ground conduit to connect the below grade conduit to the equipment control box; wiring to land the control cables on the terminal blocks in the equipment control box; jumpers and fittings to connect to substation bus. Steel stand material cost is the cost to procure and deliver one steel stand for a voltage transformer to site (laydown yard). Steel stand installation cost is the cost to place each steel stand on the foundation. Foundation size is the amount of cubic yards of concrete required for the foundation. Foundation cost is the combination of the material and installation cost for the foundation. The unit costs are inclusive of all applicable taxes.

Voltage transformer (set of 3) design details									
Design detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Foundation size	As shown	Cu. Yd	1.8	2.3	2.7	3.4	4.0	8.0	12.1

Voltage transformer (set of 3) unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	Set of 3	\$38,302	\$47,878	\$56,396	\$62,525	\$66,635	\$82,072	\$128,197
Installation	1	Set of 3	\$17,236	\$21,545	\$25,378	\$28,136	\$29,986	\$36,932	\$57,689
Steel stand material	1	Set of 3	\$1,502	\$1,876	\$2,153	\$2,424	\$2,696	\$5,381	\$7,534
Steel stand installation	1	Set of 3	\$1,727	\$2,157	\$2,475	\$2,788	\$3,100	\$6,188	\$8,664
Foundation	1	Set of 3	\$2,304	\$2,944	\$3,456	\$4,352	\$5,120	\$10,240	\$15,488
Total cost	1	Set of 3	\$61,071	\$76,400	\$89,857	\$100,225	\$107,537	\$140,814	\$217,571

4.2.2.4 Current Transformer(s)

This unit cost is the cost associated with one set of three current transformers. Material cost is the cost to procure and deliver one set of three current transformers to site (laydown yard). Installation cost includes: assembly; placement on the steel stand; connect to main ground grid; above ground conduit to connect the below grade conduit to the equipment control box; wiring to land the control cables on the terminal blocks in the equipment control box; jumpers and fittings to connect to substation bus. Steel stand material cost is the cost to procure and deliver one steel stand to site (laydown yard). Steel stand installation cost is the cost to place each steel stand on the foundation. Foundation size is the amount of cubic yards of concrete required for the foundation. Foundation cost is the combination of the material and installation cost for the foundation. The unit costs are inclusive of all applicable taxes.

Current transformer (set of 3) design details									
Design detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Foundation Size	As shown	Cu. Yd	1.8	2.3	2.7	3.4	4.0	8.0	12.1

Current transformer (set of 3) unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	Set of 3	\$60,221	\$75,276	\$102,521	\$112,791	\$123,062	\$205,103	\$358,914
Installation	1	Set of 3	\$15,055	\$18,819	\$25,630	\$28,198	\$30,765	\$51,276	\$89,729
Steel stand material	1	Set of 3	\$1,502	\$1,876	\$2,153	\$2,424	\$2,696	\$5,381	\$7,534
Steel stand installation	1	Set of 3	\$1,727	\$2,157	\$2,475	\$2,788	\$3,100	\$6,188	\$8,664
Foundation	1	Set of 3	\$2,304	\$2,944	\$3,456	\$4,352	\$5,120	\$10,240	\$15,488
Total cost	1	Set of 3	\$80,808	\$101,072	\$136,235	\$150,553	\$164,743	\$278,188	\$480,328

4.2.2.5 Bus Support, Bus and Fittings

This unit cost is the cost associated with one three-phase bus support, and its associated bus and fittings. Material cost is the cost to procure and deliver to site (laydown yard) a set of three: insulators; electrical aluminum bus; all required bus fittings. Installation cost includes: assembly; electrically connect bus (as supported by this bus stand) to adjacent electrical equipment. Steel stand material cost is the cost to procure and deliver one steel stand to site (laydown yard). Steel stand installation cost is the cost to place each steel stand on the foundation. Foundation size is the amount of cubic yards of concrete required for the foundation. Foundation cost is the combination of the material and installation cost for the foundation. The unit costs are inclusive of all applicable taxes.

Bus support, bus, and fittings (3-phase) design details									
Design detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Foundation size	As shown	Cu. Yd	3.1	3.9	4.8	6.0	7.2	9.6	14.4

Bus Support, Bus, and Fittings (3-phase) unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	3-phase set	\$5,617	\$7,021	\$8,118	\$8,523	\$8,928	\$10,558	\$12,167
Installation	1	3-phase set	\$6,740	\$8,426	\$9,742	\$10,227	\$10,713	\$12,669	\$14,600
Steel stand material	1	3-phase set	\$3,008	\$3,762	\$4,305	\$4,843	\$5,381	\$10,773	\$15,078
Steel stand installation	1	3-phase set	\$3,460	\$4,326	\$4,951	\$5,570	\$6,188	\$12,389	\$17,339
Foundation	1	3-phase set	\$3,968	\$4,992	\$6,144	\$7,680	\$9,216	\$12,288	\$18,432
Total cost	1	3-phase set	\$22,793	\$28,527	\$33,259	\$36,843	\$40,427	\$58,676	\$77,616

4.2.2.6 Deadend(s)

This unit cost is the cost associated with one full-tension deadend structure. The unit cost for a deadend structure installed in a substation is shown in the transmission line section 4.1.2.5. The same unit cost is used for substation estimates as is used for transmission line estimates.

4.2.2.7 Power Transformers

This unit cost is the cost associated with one power transformer. Power transformer cost varies based on the low side voltage winding and high side voltage winding. Unit cost includes all material, shipping, foundation, and installation costs with that transformer. The unit cost is inclusive of all applicable taxes.

Power transformer (\$/MVA)							
Winding voltage level	69kV	115kV	138kV	161kV	230kV	345kV	500kV
69kV	\$4,603	\$3,749	\$4,154	\$4,373	\$4,845	\$5,948	\$7,668
115kV	\$3,749	\$5,100	\$4,154	\$4,373	\$4,845	\$5,651	\$6,938
138kV	\$4,154	\$4,154	\$5,651	\$4,603	\$4,845	\$5,651	\$6,938
161kV	\$4,373	\$4,373	\$4,603	\$6,262	\$5,100	\$5,948	\$7,303
230kV	\$4,845	\$4,845	\$4,845	\$5,100	\$6,938	\$5,948	\$7,303
345kV	\$5,948	\$5,651	\$5,651	\$5,948	\$5,948	\$8,454	\$7,668
500kV	\$7,668	\$6,938	\$6,938	\$7,303	\$7,303	\$7,668	\$11,330

4.2.2.8 Reactors

This unit cost is the cost associated with one reactor. Reactor cost is calculated based on a \$/MVA cost. Unit cost includes all material, shipping, foundation, and installation costs with that reactor. The unit cost is inclusive of all applicable taxes.

Reactor (\$/MVA)							
Voltage Class	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Subtotal	\$13,248	\$13,248	\$13,248	\$13,248	\$13,248	\$13,248	\$13,248

4.2.2.9 Capacitor Bank

This unit cost is the cost associated with one capacitor bank. Capacitor bank costs are calculated based on a \$/MVar cost. Unit cost includes all material, shipping, foundation, and installation costs with that capacitor bank. The unit cost is inclusive of all applicable taxes.

Capacitor Bank (\$/MVar)							
Voltage Class	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Subtotal	\$9,738	\$9,738	\$9,738	\$9,738	\$9,738	\$9,738	\$9,738

4.2.2.10 Static VAr Compensators

This unit cost is for the cost associated with one static VAr compensator. Static VAr compensators costs are calculated based on a \$/MVar cost. Unit cost includes all material, shipping, foundation, and installation costs with that static VAr compensator. The unit cost is inclusive of all applicable taxes.

Static Var Compensator (\$/MVar)							
Voltage Class	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Subtotal	\$93,829	\$93,829	\$93,829	\$93,829	\$93,829	\$93,829	\$93,829

4.2.3 Protection and Control

4.2.3.1 Control Enclosure

This unit cost is the cost associated with one control enclosure. Material cost is the cost to procure and deliver one control enclosure to site (laydown yard). Procurement of the control enclosure includes all components (e.g. AC panels, DC panels, cable tray, etc) and includes all internal wiring. Relay Panels are considered in a separate unit cost. Installation cost includes: offload and placement of the control enclosure on the foundation; wiring of the AC/DC systems to field equipment; wiring of station service power; and wiring of control enclosure to communication systems. Foundation size is the amount of cubic yards of concrete required for the foundation. Foundation cost is the combination of the material and installation cost for the foundation. The unit costs are inclusive of all applicable taxes.

Control enclosure foundation design details									
Design detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Foundation size	As shown	Cu. Yd	18.0	18.0	18.0	18.0	18.0	18.0	18.0

Control enclosure unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	EA	\$264,266	\$264,266	\$264,266	\$264,266	\$264,266	\$264,266	\$264,266
Installation	1	EA	\$301,535	\$301,535	\$301,535	\$301,535	\$301,535	\$301,535	\$301,535
Foundation	1	EA	\$23,040	\$23,040	\$23,040	\$23,040	\$23,040	\$23,040	\$23,040
Total cost	1	EA	\$588,840	\$588,840	\$588,840	\$588,840	\$588,840	\$588,840	\$588,840

4.2.3.2 Relay Panel

This unit cost is the cost associated with one relay panel. Material cost is the cost to procure and deliver one relay panel to site (laydown yard). Procurement of the relay panel includes all the relays and devices in the panel, and all the internal wiring for the devices in each individual relay panel. Installation cost includes: placement of relay panel in control enclosure; wiring from field equipment; inter-panel wiring to other relay panels inside control enclosure. The unit costs are inclusive of all applicable taxes.

Relay panel unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	EA	\$18,245	\$22,806	\$28,577	\$32,149	\$35,721	\$47,632	\$59,542
Installation	1	EA	\$36,490	\$45,613	\$57,154	\$64,298	\$71,443	\$95,264	\$119,085
Total cost	1	EA	\$54,735	\$68,419	\$85,731	\$96,447	\$107,164	\$142,895	\$178,627

4.2.3.3 Conduit

This unit cost is the cost associated with 1000 feet of conduit. Material cost is the cost to procure and deliver 1000 feet of conduit to site (laydown yard). Included in the material cost is the conduit along with applicable fittings and connectors. Installation cost includes excavation, placement of conduit, and utilizing all applicable fittings and connectors. The unit costs are inclusive of all applicable taxes.

Conduit unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1000	feet	\$3,500	\$3,500	\$3,500	\$3,500	\$3,500	\$3,500	\$3,500
Installation	1000	feet	\$1,750	\$1,750	\$1,750	\$1,750	\$1,750	\$1,750	\$1,750
Total cost	1000	feet	\$5,250	\$5,250	\$5,250	\$5,250	\$5,250	\$5,250	\$5,250

4.2.3.4 Control Cable

This unit cost is the cost associated with 1000 feet of control cable. Material cost is the cost to procure and deliver 1000 feet of control cable to site (laydown yard). Installation cost includes placing, and pulling control cable in conduit and/or cable trench, and bringing the control cable to its end point where it will be landed. Final wiring of landing on terminal blocks is included in other unit costs. The unit costs are inclusive of all applicable taxes.

Control cable unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1000	feet	\$3,000	\$3,000	\$3,000	\$3,000	\$3,000	\$4,000	\$4,000
Installation	1000	feet	\$1,500	\$1,500	\$1,500	\$1,500	\$1,500	\$1,500	\$1,500
Total cost	1000	feet	\$4,500	\$4,500	\$4,500	\$4,500	\$4,500	\$5,500	\$5,500

4.2.3.5 Cable Trench

This unit cost is the cost associated with 1 foot of cable trench inclusive of lid/cover. Material cost is the cost to procure and deliver 1 foot of cable trench to site (laydown yard). Installation cost includes excavation, and placement of cable trench. Placement of control cables in cable trench is included in the control cable installation cost. The unit costs are inclusive of all applicable taxes.

Cable trench unit costs									
Unit detail	Unit	Unit of measurement	69kV	115kV	138kV	161kV	230kV	345kV	500kV
Material	1	foot	\$200	\$200	\$200	\$200	\$200	\$200	\$200
Installation	1	foot	\$50	\$50	\$50	\$50	\$50	\$50	\$50
Total cost	1	foot	\$250	\$250	\$250	\$250	\$250	\$250	\$250

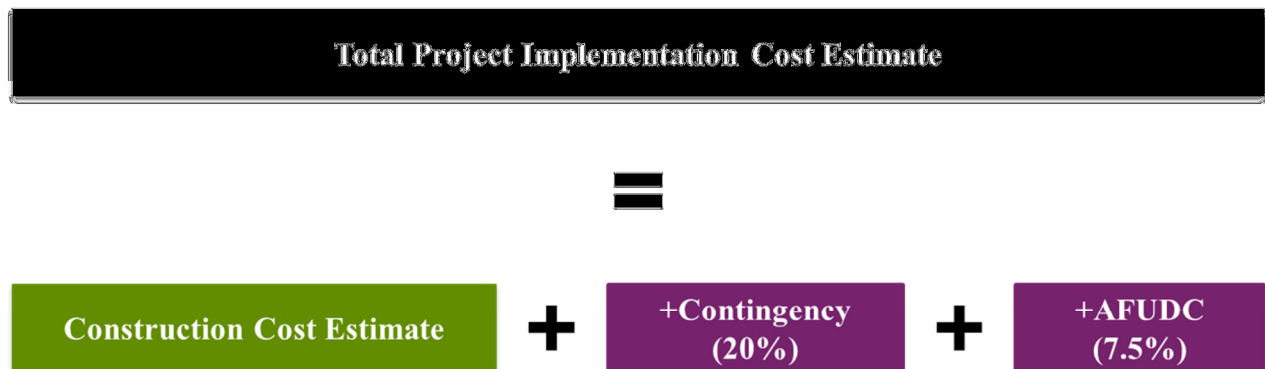
4.2.4 Overhead and professional services

4.2.4.1 Overhead and professional services

Overhead and professional services encompass all the indirect overhead costs (e.g. computers, trucks) and all the professional services (e.g. surveying, drafting, engineering, project management, testing and commissioning) required for potential projects. This cost adder is to account for all the added costs that are spread out over the entire project and are not easily quantifiable by individual items. MISO aggregates all the overhead and professionals services into one unit cost in order to facilitate discussion on the larger cost adder value that MISO considers for these costs. The contributions of these services are at a cost of 10% of the construction costs for the project.

5. Project Implementation Cost

MISO's final cost estimate dollar value provided is for the total project implementation cost of the potential project. The project implementation cost is inclusive of the construction cost (including overhead and professional services), contingency, and Allowance for Funds Used During Construction (AFUDC). The construction cost is the scope of work of the project as defined in the solution idea. Contingency is a 20% cost adder to account for all the unknowns and level of scope definition at the time of estimation. AFUDC is a 7.5% of construction cost plus contingency and added to total cost to account for the cost of debt and/or the cost of equity required to develop the project. As shown in the diagram below, construction cost plus contingency plus AFUDC provides the total project implementation cost estimate that MISO uses for its project cost estimates.



6. Exploratory Cost Estimates

MISO provides exploratory cost estimates for projects with low level of scope definition and to help screen numerous solution ideas not targeted for congestion in particular region. The exploratory cost estimates in section 6 of MISO's transmission and substation project cost estimation guide are intended to be a very high-level cost estimate for potential projects located in MISO footprint. MISO does not recommend using these exploratory unit costs for any particular solution idea in the regular planning cycle due to the breadth of the assumptions used to derive the unit costs and lower level of granularity regarding specific project components. The exploratory cost estimates provided below are based on MISO's 2018 cost estimation assumptions and cost data. The costs presented in section 6 include contingency and AFUDC. The costs presented in section 6 are in 2018 dollars.

6.1 Transmission Line exploratory costs

Single circuit transmission line (\$/Mile)							
Location – state	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Arkansas	\$2.0M	\$2.2M	\$2.4M	\$2.5M	\$2.8M	\$4.2M	\$4.6M
Illinois	\$1.6M	\$1.7M	\$1.8M	\$1.9M	\$2.1M	\$3.2M	\$3.4M
Indiana	\$1.5M	\$1.6M	\$1.8M	\$1.8M	\$2.0M	\$3.1M	\$3.3M
Iowa	\$1.4M	\$1.6M	\$1.7M	\$1.7M	\$1.9M	\$2.9M	\$3.1M
Kentucky	\$1.5M	\$1.7M	\$1.8M	\$1.8M	\$2.0M	\$3.1M	\$3.3M
Louisiana	\$2.1M	\$2.3M	\$2.5M	\$2.6M	\$2.9M	\$4.4M	\$4.8M
Michigan	\$1.5M	\$1.7M	\$1.8M	\$1.8M	\$2.0M	\$3.1M	\$3.3M
Minnesota	\$1.5M	\$1.7M	\$1.8M	\$1.9M	\$2.1M	\$3.2M	\$3.4M
Mississippi	\$2.1M	\$2.3M	\$2.5M	\$2.6M	\$2.9M	\$4.4M	\$4.7M
Missouri	\$1.5M	\$1.6M	\$1.8M	\$1.8M	\$2.0M	\$3.1M	\$3.3M
Montana	\$1.3M	\$1.4M	\$1.6M	\$1.6M	\$1.7M	\$2.7M	\$2.8M
North Dakota	\$1.3M	\$1.4M	\$1.6M	\$1.6M	\$1.7M	\$2.7M	\$2.8M
South Dakota	\$1.3M	\$1.5M	\$1.6M	\$1.6M	\$1.7M	\$2.7M	\$2.9M
Texas	\$1.8M	\$1.9M	\$2.1M	\$2.1M	\$2.4M	\$3.6M	\$3.9M
Wisconsin	\$1.7M	\$1.8M	\$2.0M	\$2.0M	\$2.2M	\$3.4M	\$3.6M

Includes contingency, and AFUDC.

Double circuit transmission line (\$/Mile)							
Location – State	69kV line	115kV line	138kV line	161kV line	230kV line	345kV line	500kV line
Arkansas	\$2.5M	\$2.8M	\$3M	\$3.1M	\$3.8M	\$5.9M	\$6.1M
Illinois	\$2.0M	\$2.2M	\$2.4M	\$2.5M	\$3.0M	\$4.7M	\$4.8M
Indiana	\$2.0M	\$2.2M	\$2.4M	\$2.4M	\$2.9M	\$4.6M	\$4.7M
Iowa	\$1.9M	\$2.1M	\$2.3M	\$2.3M	\$2.8M	\$4.4M	\$4.5M
Kentucky	\$2.0M	\$2.2M	\$2.4M	\$2.5M	\$2.9M	\$4.7M	\$4.8M
Louisiana	\$2.6M	\$2.9M	\$3.1M	\$3.2M	\$3.9M	\$6.1M	\$6.3M
Michigan	\$2.0M	\$2.2M	\$2.4M	\$2.5M	\$2.9M	\$4.7M	\$4.8M
Minnesota	\$2.0M	\$2.3M	\$2.5M	\$2.5M	\$3.0M	\$4.8M	\$4.9M
Mississippi	\$2.6M	\$2.9M	\$3.1M	\$3.2M	\$3.9M	\$6.0M	\$6.3M
Missouri	\$2.0M	\$2.2M	\$2.4M	\$2.4M	\$2.9M	\$4.6M	\$4.7M
Montana	\$1.8M	\$2.0M	\$2.2M	\$2.2M	\$2.7M	\$4.3M	\$4.3M
North Dakota	\$1.8M	\$2.0M	\$2.2M	\$2.2M	\$2.7M	\$4.3M	\$4.3M
South Dakota	\$1.8M	\$2.0M	\$2.2M	\$2.2M	\$2.7M	\$4.3M	\$4.3M
Texas	\$2.3M	\$2.5M	\$2.7M	\$2.8M	\$3.4M	\$5.3M	\$5.5M
Wisconsin	\$2.1M	\$2.4M	\$2.6M	\$2.6M	\$3.2M	\$5.0M	\$5.1M

Includes contingency, and AFUDC.

6.2 Substation exploratory costs

Where the substation site is existing, MISO assumes the following exploratory cost to add one line position to an existing substation. This exploratory cost is not site specific, and assumes that the substation site has to be physically expanded, and that new bus work has to be installed.

Substation upgrade - add one line position							
Voltage class	69kV	115kV	138kV	161kV	230kV	345kV	500kV
All states	\$1.6M	\$1.9M	\$2.3M	\$2.6M	\$3.1M	\$5.4M	\$8.8M

Includes contingency, and AFUDC.

Where the substation site is new, MISO assumes the following exploratory cost estimate where the new site has four line positions, and a breaker-and-a-half bus arrangement. The number of line positions and bus arrangement were selected to be indicative of a potential new substation layout.

New substation site (4 new line positions and breaker-and-a-half arrangement)							
Voltage class	69kV	115kV	138kV	161kV	230kV	345kV	500kV
All states	\$5.7M	\$6.8M	\$7.8M	\$8.9M	\$10.8M	\$18.7M	\$31.4M

Includes contingency, and AFUDC.

6.3 Add one power transformer to an existing substation site

Where a power transformer is to be added to an existing substation, MISO has created the exploratory cost below. The exploratory cost varies based on the primary and secondary winding voltages of the power transformer. Additionally, the exploratory cost estimate includes the cost of site work, equipment, protection and control, and overhead and professional services unit costs that would be required to interconnection the power transformer into an existing substation site.

Add one 700MVA power transformer to an existing substation site							
Winding voltage level	69kV	115kV	138kV	161kV	230kV	345kV	500kV
69kV	\$7.8M	\$7.2M	\$7.8M	\$8.2M	\$9.5M	\$11.7M	\$15.7M
115kV	\$7.2M	\$8.7M	\$8.0M	\$8.4M	\$9.7M	\$11.6M	\$15.1M
138kV	\$7.8M	\$8.0M	\$9.8M	\$8.9M	\$9.9M	\$11.8M	\$15.4M
161kV	\$8.2M	\$8.4M	\$8.9M	\$10.7M	\$10.3M	\$12.3M	\$15.9M
230kV	\$9.5M	\$9.7M	\$9.9M	\$10.3M	\$13.0M	\$13.1M	\$16.7M
345kV	\$11.7M	\$11.6M	\$11.8M	\$12.3M	\$13.1M	\$16.6M	\$18.1M
500kV	\$15.7M	\$15.1M	\$15.4M	\$15.9M	\$16.7M	\$18.1M	\$24.1M

Includes contingency, and AFUDC.

Add one 1200MVA power transformer to an existing substation site							
Winding voltage level	69kV	115kV	138kV	161kV	230kV	345kV	500kV
69kV	\$11.1M	\$9.9M	\$10.8M	\$11.3M	\$12.9M	\$15.9M	\$21.1M
115kV	\$9.9M	\$12.4M	\$11M	\$11.5M	\$13.1M	\$15.6M	\$20.1M
138kV	\$10.8M	\$11.0M	\$13.8M	\$12.1M	\$13.4M	\$15.8M	\$20.3M
161kV	\$11.3M	\$11.5M	\$12.1M	\$15.1M	\$14.0M	\$16.5M	\$21.1M
230kV	\$12.9M	\$13.1M	\$13.4M	\$14.0M	\$17.9M	\$17.3M	\$21.9M
345kV	\$15.9M	\$15.6M	\$15.8M	\$16.5M	\$17.3M	\$22.6M	\$23.6M
500kV	\$21.1M	\$20.1M	\$20.3M	\$21.1M	\$21.9M	\$23.6M	\$32.1M

Includes contingency, and AFUDC.

Add one 2250MVA power transformer to an existing substation site							
Winding Voltage level	69kV	115kV	138kV	161kV	230kV	345kV	500kV
69kV	\$18.0M	\$15.4M	\$17.0M	\$17.8M	\$20.1M	\$24.7M	\$32.5M
115kV	\$15.4M	\$20.0M	\$17.2M	\$18.0M	\$20.3M	\$24.0M	\$30.4M
138kV	\$17.0M	\$17.2M	\$22.2M	\$19.0M	\$20.6M	\$24.2M	\$30.6M
161kV	\$17.8M	\$18.0M	\$19.0M	\$24.5M	\$21.6M	\$25.3M	\$32.0M
230kV	\$20.1M	\$20.3M	\$20.6M	\$21.6M	\$28.2M	\$26.1M	\$32.8M
345kV	\$24.7M	\$24.0M	\$24.2M	\$25.3M	\$26.1M	\$35.2M	\$35.0M
500kV	\$32.5M	\$30.4M	\$30.6M	\$32.0M	\$32.8M	\$35.0M	\$49.0M

Includes contingency, and AFUDC.